

Assignment-9

Goal: To perform RNA-seq analysis using two different aligners, HISAT2 and STAR, followed by differential expression and downstream analyses.

Materials needed:

- RNA-seq FASTQ files for the samples: SRR15276765, SRR15276766, SRR15276767, SRR15276773
- *Plasmodium falciparum* reference genome.fasta & annotation.gtf files
- FastQC
- Trim_galore!
- HISAT2
- Qualimap
- MultiQC

Sub-directories to be created:

- 1_Raw_Data
- 2_Reference
- 3_FastQC
- 4_TrimGalore
- 5_HiSAT2
- 6_STAR

Exercise 1: RNA-seq Read Mapping and Quality Assessment using hisat2

1. Data Download:

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug$ cd 1_Raw_Data/
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/1_Raw_Data$ wget ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR152/073/SRR15276773/SRR15276773_1.fastq.gz
--2023-08-24 23:45:35-- ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR152/073/SRR15276773/SRR15276773_1.fastq.gz
=> 'SRR15276773_1.fastq.gz'
Resolving ftp.sra.ebi.ac.uk (ftp.sra.ebi.ac.uk)... 193.62.193.165
Connecting to ftp.sra.ebi.ac.uk (ftp.sra.ebi.ac.uk)[193.62.193.165]:21... connected.
```

Similarly, download other samples also using 'wget' command.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/1_Raw_Data$ ls -lh
total 5.1G
-rw-r--r-- 1 ibab ibab 176M Aug 25 00:05 SRR15276765_1.fastq.gz
-rw-r--r-- 1 ibab ibab 184M Aug 25 00:06 SRR15276765_2.fastq.gz
-rw-r--r-- 1 ibab ibab 780M Aug 25 00:11 SRR15276766_1.fastq.gz
-rw-r--r-- 1 ibab ibab 727M Aug 25 00:19 SRR15276766_2.fastq.gz
-rw-r--r-- 1 ibab ibab 879M Aug 25 00:27 SRR15276767_1.fastq.gz
-rw-r--r-- 1 ibab ibab 818M Aug 25 00:35 SRR15276767_2.fastq.gz
-rw-r--r-- 1 ibab ibab 832M Aug 24 23:56 SRR15276773_1.fastq.gz
-rw-r--r-- 1 ibab ibab 795M Aug 25 00:04 SRR15276773_2.fastq.gz
```

Now, download the reference genome (.fasta) and annotation (.gtf) files of *Plasmodium falciparum*.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug$ cd 2_Reference/
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/2_Reference$ ls -lh
total 27M
-rw-r--r-- 1 ibab ibab 23M Aug 25 00:39 Pfalciparum.genome.fasta
-rw-r--r-- 1 ibab ibab 3.7M Aug 25 00:40 Pfalciparum.genome.gtf
```

2. Preprocessing:

Run FastQC and TrimGalore followed by another round of FastQC on the .val files.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug$ cd 3_Fastqc/
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/3_Fastqc$ fastqc /home/ibab/NGS/23aug/1_Raw_Data/*.gz -o /home/ibab/NGS/23aug/3_Fastqc
```

```
Started analysis of SRR15276765_1.fastq.gz
Approx 5% complete for SRR15276765_1.fastq.gz
Approx 10% complete for SRR15276765_1.fastq.gz
Approx 15% complete for SRR15276765_1.fastq.gz
Approx 20% complete for SRR15276765_1.fastq.gz
Approx 25% complete for SRR15276765_1.fastq.gz
Approx 30% complete for SRR15276765_1.fastq.gz
Approx 35% complete for SRR15276765_1.fastq.gz
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/3_Fastqc$ ls -lh
total 7.9M
-rw-r--r-- 1 ibab ibab 690K Aug 25 00:53 SRR15276765_1_fastqc.html
-rw-r--r-- 1 ibab ibab 341K Aug 25 00:53 SRR15276765_1_fastqc.zip
-rw-r--r-- 1 ibab ibab 691K Aug 25 00:53 SRR15276765_2_fastqc.html
-rw-r--r-- 1 ibab ibab 349K Aug 25 00:53 SRR15276765_2_fastqc.zip
-rw-r--r-- 1 ibab ibab 637K Aug 25 00:55 SRR15276766_1_fastqc.html
-rw-r--r-- 1 ibab ibab 355K Aug 25 00:55 SRR15276766_1_fastqc.zip
-rw-r--r-- 1 ibab ibab 647K Aug 25 00:56 SRR15276766_2_fastqc.html
-rw-r--r-- 1 ibab ibab 345K Aug 25 00:56 SRR15276766_2_fastqc.zip
-rw-r--r-- 1 ibab ibab 636K Aug 25 00:58 SRR15276767_1_fastqc.html
-rw-r--r-- 1 ibab ibab 348K Aug 25 00:58 SRR15276767_1_fastqc.zip
-rw-r--r-- 1 ibab ibab 647K Aug 25 00:59 SRR15276767_2_fastqc.html
-rw-r--r-- 1 ibab ibab 344K Aug 25 00:59 SRR15276767_2_fastqc.zip
-rw-r--r-- 1 ibab ibab 636K Aug 25 01:01 SRR15276773_1_fastqc.html
-rw-r--r-- 1 ibab ibab 344K Aug 25 01:01 SRR15276773_1_fastqc.zip
-rw-r--r-- 1 ibab ibab 648K Aug 25 01:03 SRR15276773_2_fastqc.html
-rw-r--r-- 1 ibab ibab 342K Aug 25 01:03 SRR15276773_2_fastqc.zip
```

```
#!/bin/bash

### This script trims paired-end FASTQ files using TrimGalore
### It allows the user to set quality cutoff and stringency, or fall back to defaults.
### It also runs FastQC on the trimmed files.

base_dir="$HOME/NGS/23aug"
raw="$base_dir/1_Raw_Data"
out="$base_dir/4_TrimGalore"
trimgalore="$HOME/NGS/Packages/TrimGalore/trim_galore"

read -p "Enter quality cutoff (default = 25): " quality
if [[ -z "$quality" ]]; then
    quality=25
fi

read -p "Enter stringency level (default = 5): " stringency
if [[ -z "$stringency" ]]; then
    stringency=5
fi

read -p "Enter sample ID(s) separated by space: " -a sample_ids

for sample in "${sample_ids[@]}; do
    # Define paths to paired-end FASTQ files
    r1="$raw/${sample}_1.fastq.gz"
    r2="$raw/${sample}_2.fastq.gz"
    # Check if both files exist
    if [[ -f "$r1" && -f "$r2" ]]; then
        # Create output subdirectory for this sample
        sample_out="$out/$sample"
        mkdir -p "$sample_out"
        echo "Running TrimGalore for $sample (quality = $quality, stringency = $stringency)"
        "$trimgalore" --paired "$r1" "$r2" -q "$quality" --stringency "$stringency" --fastqc -o "$sample_out"
    else
        echo "One or both FASTQ files not found for sample: $sample"
    fi
done
```

```

ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/1_Raw_Data$ ./trim_with_fastqc.sh
Enter quality cutoff (default = 25): 28
Enter stringency level (default = 5): 5
Enter sample ID(s) separated by space: SRR15276765 SRR15276766 SRR15276767 SRR15276773
Running TrimGalore for SRR15276765 (quality = 28, stringency = 5)
Multicore support not enabled. Proceeding with single-core trimming.
Path to Cutadapt set as: 'cutadapt' (default)
Cutadapt seems to be working fine (tested command 'cutadapt --version')
Cutadapt version: 3.5
single-core operation.
Proceeding with 'gzip' for decompression
To decrease CPU usage of decompression, please install 'igzip' and run again

No quality encoding type selected. Assuming that the data provided uses Sanger encoded Phred scores (default)

Output will be written into the directory: /home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/

AUTO-DETECTING ADAPTER TYPE
=====
Attempting to auto-detect adapter type from the first 1 million sequences of the first file (>> /home/ibab/NGS/23aug/1_Raw_Data/SRR15276765_1.fastq.gz <<)

Found perfect matches for the following adapter sequences:
Adapter type   Count   Sequence   Sequences analysed   Percentage
Illumina       83      AGATCGGAAGAGC   1000000   0.01
Nextera 7      CTGTCTCTTATA   1000000   0.00
smallRNA       2       TGAATTCTCGG    1000000   0.00
Using Illumina adapter for trimming (count: 83). Second best hit was Nextera (count: 7)

Writing report to '/home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/SRR15276765_1.fastq.gz_trimming_report.txt'

SUMMARISING RUN PARAMETERS
=====
Input filename: /home/ibab/NGS/23aug/1_Raw_Data/SRR15276765_1.fastq.gz
Trimming mode: paired-end
Trim Galore version: 0.6.10
Cutadapt version: 3.5
Number of cores used for trimming: 1
Quality Phred score cutoff: 28
Quality encoding type selected: ASCII+33
Adapter sequence: 'AGATCGGAAGAGC' (Illumina TruSeq, Sanger iPCR; auto-detected)

ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug$ cd 4_TrimGalore/
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/4_TrimGalore$ ls -lh
total 16K
drwxr-xr-x 2 ibab ibab 4.0K Aug 25 01:34 SRR15276765
drwxr-xr-x 2 ibab ibab 4.0K Aug 25 06:22 SRR15276766
drwxr-xr-x 2 ibab ibab 4.0K Aug 25 06:43 SRR15276767
drwxr-xr-x 2 ibab ibab 4.0K Aug 25 07:02 SRR15276773
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/4_TrimGalore$ cd SRR15276765
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/4_TrimGalore/SRR15276765$ ls -lh
total 338M
-rw-r--r-- 1 ibab ibab 2.8K Aug 25 01:29 SRR15276765_1.fastq.gz_trimming_report.txt
-rw-r--r-- 1 ibab ibab 166M Aug 25 01:33 SRR15276765_1_val_1.fq.gz
-rw-r--r-- 1 ibab ibab 664K Aug 25 01:34 SRR15276765_1_val_1_fastqc.html
-rw-r--r-- 1 ibab ibab 345K Aug 25 01:34 SRR15276765_1_val_1_fastqc.zip
-rw-r--r-- 1 ibab ibab 2.8K Aug 25 01:33 SRR15276765_2.fastq.gz_trimming_report.txt
-rw-r--r-- 1 ibab ibab 170M Aug 25 01:33 SRR15276765_2_val_2.fq.gz
-rw-r--r-- 1 ibab ibab 663K Aug 25 01:34 SRR15276765_2_val_2_fastqc.html
-rw-r--r-- 1 ibab ibab 350K Aug 25 01:34 SRR15276765_2_val_2_fastqc.zip

```

SRR15276765_1

Before trimming

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ⚠ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ⚠ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✓ [Adapter Content](#)

After trimming

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ⚠ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ⚠ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✓ [Adapter Content](#)

SRR15276765_2

Before trimming

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ⚠ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ⚠ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✓ [Adapter Content](#)

After trimming

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ⚠ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ⚠ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✓ [Adapter Content](#)

3. Reference Indexing

The hisat2-build step is essential before aligning sequencing reads, because HISAT2 can't directly use the plain FASTA genome sequence for mapping. Instead, it converts the reference genome into a specialized set of index files that encode the sequence in a compressed and searchable form using algorithms like the Burrows–Wheeler Transform (BWT) and FM-index. These index files allow HISAT2 to very quickly find the exact or approximate location of each read in the genome without scanning through the entire sequence every time, which saves a huge amount of time and memory during alignment. Once built, the index can be reused multiple times for different experiments or datasets, so we don't need to repeat this step unless the reference genome changes.

```
cd 5_HiSAT2
```

```
mkdir 1_Index
```

```
mkdir 2_Alignment
```

```
cd 1_Index
```

```
hisat2-build /home/ibab/NGS/23aug/2_Reference/Pfalcciparum.genome.fasta  
Pfalcciparum.genome
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/1_Index$ hisat2-build /home/ibab/NGS/23aug/2_Reference/Pfalcciparum.genome.fasta Pfalcciparum.genome
Settings:
  Output files: "Pfalcciparum.genome.*.ht2"
  Line rate: 6 (line is 64 bytes)
  Lines per side: 1 (side is 64 bytes)
  Offset rate: 4 (one in 16)
  FTable chars: 10
  Strings: unpacked
  Local offset rate: 3 (one in 8)
  Local fTable chars: 6
  Local sequence length: 57344
```

Indexing the reference genome using HISAT2 creates eight files with extensions from .ht1 to .ht8

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/1_Index$ ls -lh
total 39M
-rw-r--r-- 1 ibab ibab 12M Aug 29 10:05 Pfalcciparum.genome.1.ht2
-rw-r--r-- 1 ibab ibab 5.6M Aug 29 10:05 Pfalcciparum.genome.2.ht2
-rw-r--r-- 1 ibab ibab 152 Aug 29 10:05 Pfalcciparum.genome.3.ht2
-rw-r--r-- 1 ibab ibab 5.6M Aug 29 10:05 Pfalcciparum.genome.4.ht2
-rw-r--r-- 1 ibab ibab 9.9M Aug 29 10:05 Pfalcciparum.genome.5.ht2
-rw-r--r-- 1 ibab ibab 5.7M Aug 29 10:05 Pfalcciparum.genome.6.ht2
-rw-r--r-- 1 ibab ibab 12 Aug 29 10:05 Pfalcciparum.genome.7.ht2
-rw-r--r-- 1 ibab ibab 8 Aug 29 10:05 Pfalcciparum.genome.8.ht2
```

4. Align val.fastq files to the reference genome using HiSAT2

```
mkdir 2_Alignment (in the same directory – 5_HiSAT2)
```

```
cd 2_Alignment
```

```
hisat2 -x /home/ibab/NGS/23aug/5_HiSAT2/1_Index/Pfalcciparum.genome -1  
/home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/SRR15276765_1_val_1.fq.gz -2
```

/home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/SRR15276765_2_val_2.fq.gz -S
SRR15276765.sam

This command is aligning paired-end RNA-seq reads from the sample SRR15276765 against the *Plasmodium falciparum* reference genome using HISAT2. The reference genome index is provided with -x, and the cleaned forward (-1) and reverse (-2) reads are taken as input from Trim Galore output FASTQ files. HISAT2 processes these reads, finds their best matching locations in the genome, and writes the results in SAM format, which can later be converted to BAM, sorted, and used for downstream analysis like transcript assembly or variant calling.

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ hisat2 -x /home/ibab/NGS/23aug/5_HiSAT2/1_Index/Pfalciparum.genome -1 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/SRR15276765_1_val_1.fq.gz -2 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276765/SRR15276765_2_val_2.fq.gz -S SRR15276765.sam
2419469 reads; of these:
  2419469 (100.00%) were paired; of these:
    297141 (12.28%) aligned concordantly 0 times
    1972793 (81.54%) aligned concordantly exactly 1 time
    149535 (6.18%) aligned concordantly >1 times
  ----
    297141 pairs aligned concordantly 0 times; of these:
      3040 (1.02%) aligned discordantly 1 time
  ----
    294101 pairs aligned 0 times concordantly or discordantly; of these:
      588202 mates make up the pairs; of these:
        549819 (93.47%) aligned 0 times
        32031 (5.45%) aligned exactly 1 time
        6352 (1.08%) aligned >1 times
88.64% overall alignment rate
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ hisat2 -x /home/ibab/NGS/23aug/5_HiSAT2/1_Index/Pfalciparum.genome -1 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276766/SRR15276766_1_val_1.fq.gz -2 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276766/SRR15276766_2_val_2.fq.gz -S SRR15276766.sam
9757129 reads; of these:
  9757129 (100.00%) were paired; of these:
    568994 (5.83%) aligned concordantly 0 times
    8904851 (91.27%) aligned concordantly exactly 1 time
    283284 (2.90%) aligned concordantly >1 times
  ----
    568994 pairs aligned concordantly 0 times; of these:
      35999 (6.33%) aligned discordantly 1 time
  ----
    532995 pairs aligned 0 times concordantly or discordantly; of these:
      1065990 mates make up the pairs; of these:
        727099 (68.21%) aligned 0 times
        309345 (29.02%) aligned exactly 1 time
        29546 (2.77%) aligned >1 times
96.27% overall alignment rate
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ hisat2 -x /home/ibab/NGS/23aug/5_HiSAT2/1_Index/Pfalciparum.genome -1 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276767/SRR15276767_1_val_1.fq.gz -2 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276767/SRR15276767_2_val_2.fq.gz -S SRR15276767.sam
11120133 reads; of these:
  11120133 (100.00%) were paired; of these:
    556502 (5.00%) aligned concordantly 0 times
    10331032 (92.90%) aligned concordantly exactly 1 time
    232599 (2.09%) aligned concordantly >1 times
  ----
    556502 pairs aligned concordantly 0 times; of these:
      47617 (8.56%) aligned discordantly 1 time
  ----
    508885 pairs aligned 0 times concordantly or discordantly; of these:
      1017770 mates make up the pairs; of these:
        617741 (60.70%) aligned 0 times
        362274 (35.59%) aligned exactly 1 time
        37755 (3.71%) aligned >1 times
97.22% overall alignment rate
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ hisat2 -x /home/ibab/NGS/23aug/5_HiSAT2/1_Index/Pfalciparum.genome -1 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276773/SRR15276773_1_val_1.fq.gz -2 /home/ibab/NGS/23aug/4_TrimGalore/SRR15276773/SRR15276773_2_val_2.fq.gz -S SRR15276773.sam
10965883 reads; of these:
  10965883 (100.00%) were paired; of these:
    478431 (4.36%) aligned concordantly 0 times
    10277895 (93.73%) aligned concordantly exactly 1 time
    209557 (1.91%) aligned concordantly >1 times
  ----
    478431 pairs aligned concordantly 0 times; of these:
      44604 (9.32%) aligned discordantly 1 time
  ----
    433827 pairs aligned 0 times concordantly or discordantly; of these:
      867654 mates make up the pairs; of these:
        484420 (55.83%) aligned 0 times
        351669 (40.53%) aligned exactly 1 time
        31565 (3.64%) aligned >1 times
97.79% overall alignment rate
```

5. Convert SAM file to BAM file for downstream analysis

```
samtools view -bS SRR15276765.sam > SRR15276765.bam
```

Since, BAM is the compressed, binary version of SAM that saves space and is faster for downstream processing.

```
samtools view -bS SRR15276766.sam > SRR15276766.bam
```

```
samtools view -bS SRR15276767.sam > SRR15276767.bam
```

```
samtools view -bS SRR15276773.sam > SRR15276773.bam
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools view -bS SRR15276765.sam > SRR15276765.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ ls -lh
total 8.8G
-rw-r--r-- 1 ibab ibab 433M Aug 29 10:38 SRR15276765.bam
-rw-r--r-- 1 ibab ibab 1.9G Aug 29 10:24 SRR15276765.sam
-rw-r--r-- 1 ibab ibab 6.6G Aug 29 10:39 SRR15276766.sam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools view -bS SRR15276766.sam > SRR15276766.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools view -bS SRR15276767.sam > SRR15276767.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools view -bS SRR15276773.sam > SRR15276773.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ ls -lh
total 32G
-rw-r--r-- 1 ibab ibab 433M Aug 29 10:38 SRR15276765.bam
-rw-r--r-- 1 ibab ibab 1.9G Aug 29 10:24 SRR15276765.sam
-rw-r--r-- 1 ibab ibab 1.8G Aug 29 10:47 SRR15276766.bam
-rw-r--r-- 1 ibab ibab 7.3G Aug 29 10:40 SRR15276766.sam
-rw-r--r-- 1 ibab ibab 2.0G Aug 29 11:05 SRR15276767.bam
-rw-r--r-- 1 ibab ibab 8.3G Aug 29 10:56 SRR15276767.sam
-rw-r--r-- 1 ibab ibab 2.0G Aug 29 11:21 SRR15276773.bam
-rw-r--r-- 1 ibab ibab 8.2G Aug 29 11:13 SRR15276773.sam
```

6. Sort the BAM file by coordinates using samtools

```
samtools sort -o SRR15276765_sorted.bam SRR15276765.bam
```

This command sorts the BAM file by genomic coordinates, which arranges the aligned reads in the order they appear along the reference genome; sorting is required for efficient indexing, visualization (e.g., in IGV), and most downstream analyses like variant calling.

```
samtools sort -o SRR15276766_sorted.bam SRR15276766.bam
```

```
samtools sort -o SRR15276767_sorted.bam SRR15276767.bam
```

```
samtools sort -o SRR15276773_sorted.bam SRR15276773.bam
```

```
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ ls -lh
total 36G
-rw-r--r-- 1 ibab ibab 433M Aug 29 10:38 SRR15276765.bam
-rw-r--r-- 1 ibab ibab 1.9G Aug 29 10:24 SRR15276765.sam
-rw-r--r-- 1 ibab ibab 277M Aug 29 11:23 SRR15276765_sorted.bam
-rw-r--r-- 1 ibab ibab 1.8G Aug 29 10:47 SRR15276766.bam
-rw-r--r-- 1 ibab ibab 7.3G Aug 29 10:40 SRR15276766.sam
-rw-r--r-- 1 ibab ibab 1015M Aug 29 11:26 SRR15276766_sorted.bam
-rw-r--r-- 1 ibab ibab 2.0G Aug 29 11:05 SRR15276767.bam
-rw-r--r-- 1 ibab ibab 8.3G Aug 29 10:56 SRR15276767.sam
-rw-r--r-- 1 ibab ibab 1.2G Aug 29 11:30 SRR15276767_sorted.bam
-rw-r--r-- 1 ibab ibab 2.0G Aug 29 11:21 SRR15276773.bam
-rw-r--r-- 1 ibab ibab 8.2G Aug 29 11:13 SRR15276773.sam
-rw-r--r-- 1 ibab ibab 1.1G Aug 29 11:36 SRR15276773_sorted.bam
```


7. Index the sorted .bam files

Indexing a BAM file creates a lookup table that tells tools where in the file to find reads for specific genome regions. Without an index, the program would need to scan the entire BAM file each time, which is very slow. With an index (.bai), we can quickly jump to a region of interest - essential for genome browsers, variant callers, and region-based analyses.

samtools index SRR15276765_sorted.bam

samtools index SRR15276766_sorted.bam

samtools index SRR15276767_sorted.bam

samtools index SRR15276773_sorted.bam

```
SRR15276765_sorted.bam SRR15276766_sorted.bam SRR15276767_sorted.bam SRR15276773_sorted.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools index SRR15276765_sorted.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools index SRR15276766_sorted.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools index SRR15276767_sorted.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ samtools index SRR15276773_sorted.bam
ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ ls -lh
total 36G
-rw-r--r-- 1 ibab ibab 433M Aug 29 10:38 SRR15276765.bam
-rw-r--r-- 1 ibab ibab 1.9G Aug 29 10:24 SRR15276765.sam
-rw-r--r-- 1 ibab ibab 277M Aug 29 11:23 SRR15276765_sorted.bam
-rw-r--r-- 1 ibab ibab 60K Sep 18 05:53 SRR15276765_sorted.bam.bai
-rw-r--r-- 1 ibab ibab 1.8G Aug 29 10:47 SRR15276766.bam
-rw-r--r-- 1 ibab ibab 7.3G Aug 29 10:40 SRR15276766.sam
-rw-r--r-- 1 ibab ibab 1015M Aug 29 11:26 SRR15276766_sorted.bam
-rw-r--r-- 1 ibab ibab 127K Sep 18 05:54 SRR15276766_sorted.bam.bai
-rw-r--r-- 1 ibab ibab 2.0G Aug 29 11:05 SRR15276767.bam
-rw-r--r-- 1 ibab ibab 8.3G Aug 29 10:56 SRR15276767.sam
-rw-r--r-- 1 ibab ibab 1.2G Aug 29 11:30 SRR15276767_sorted.bam
-rw-r--r-- 1 ibab ibab 146K Sep 18 05:54 SRR15276767_sorted.bam.bai
-rw-r--r-- 1 ibab ibab 2.0G Aug 29 11:21 SRR15276773.bam
-rw-r--r-- 1 ibab ibab 8.2G Aug 29 11:13 SRR15276773.sam
-rw-r--r-- 1 ibab ibab 1.1G Aug 29 11:36 SRR15276773_sorted.bam
-rw-r--r-- 1 ibab ibab 141K Sep 18 05:55 SRR15276773_sorted.bam.bai
```

8. Quality check of HiSAT2 alignment using Qualimap

Qualimap is used to assess the quality of sequencing read alignments stored in BAM files. It generates detailed statistics and visual reports, including mapping quality, coverage distribution, and GC content, which help evaluate the reliability of the alignment data.

qualimap bamqc -bam SRR15276765_sorted.bam -outfile SRR15276765.pdf -outformat pdf

qualimap bamqc -bam SRR15276766_sorted.bam -outfile SRR15276766.pdf -outformat pdf

qualimap bamqc -bam SRR15276767_sorted.bam -outfile SRR15276767.pdf -outformat pdf

qualimap bamqc -bam SRR15276773_sorted.bam -outfile SRR15276773.pdf -outformat pdf

This command runs **Qualimap's BAMQC module** to perform quality control on your sorted BAM file. The **option -bam** tells Qualimap which alignment file to analyze, while **-outfile** specifies the name of the QC report it should generate. The **-outformat pdf flag** ensures that the report is produced as a PDF file. In short, this command takes the aligned reads, evaluates metrics like mapping quality, coverage, GC content, and insert size, and produces a PDF report summarizing the alignment quality for downstream interpretation.


```

(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ qualimap bamqc -bam SRR15276765_sorted.bam -outfile SRR15276765.pdf -outformat pdf
Command line run, unsetting DISPLAY variable...
Display:
Java memory size is set to 1200M
Launching application...

QualiMap v.2.2.2-dev
Built on 2016-12-11 14:41

Selected tool: bamqc
Available memory (Mb): 35
Max memory (Mb): 1258
Starting bam qc....
Loading sam header...
Loading locator...
Loading reference...
Number of windows: 400, effective number of windows: 415
Chunk of reads size: 1000
Number of threads: 4
Processed 50 out of 415 windows...
Processed 100 out of 415 windows...
Processed 150 out of 415 windows...

```

SRR15276765

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	4,838,938
Mapped reads	4,289,119 / 88.64%
Unmapped reads	549,819 / 11.36%
Mapped paired reads	4,289,119 / 88.64%
Mapped reads, first in pair	2,143,726 / 44.3%
Mapped reads, second in pair	2,145,393 / 44.34%
Mapped reads, both in pair	4,261,018 / 88.06%
Mapped reads, singletons	28,101 / 0.58%
Secondary alignments	386,859
Read min/max/mean length	20 / 126 / 131.08
Duplicated reads (estimated)	2,938,054 / 60.72%
Duplication rate	54.12%
Clipped reads	510,231 / 10.54%

2.2. ACGT Content

Number/percentage of A's	188,256,000 / 36.02%
Number/percentage of C's	72,822,733 / 13.93%
Number/percentage of T's	184,587,841 / 35.31%
Number/percentage of G's	77,047,850 / 14.74%
Number/percentage of N's	152,659,450 / 29.21%
GC Percentage	28.67%

2.3. Coverage

Mean	28.9485
Standard Deviation	202.9304

2.4. Mapping Quality

Mean Mapping Quality	44.59
----------------------	-------

2.5. Insert size

Mean	361.52
Standard Deviation	7,751.45
P25/Median/P75	165 / 194 / 235

2.6. Mismatches and indels

General error rate	0.11%
Mismatches	624,159
Insertions	109,438
Mapped reads with at least one insertion	2.52%
Deletions	55,809
Mapped reads with at least one deletion	1.3%
Homopolymer indels	84.71%

2.7. Chromosome stats

Name	Length	Mapped bases	Mean coverage	Standard deviation
------	--------	--------------	---------------	--------------------

Bioinformatics - Genomics				
PF3D7_01_v3	640851	12825156	20.0127	59.4555
PF3D7_02_v3	947102	17736555	18.7272	38.716
PF3D7_03_v3	1067971	24803400	23.2248	53.2813

SRR15276766

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	19,514,258
Mapped reads	18,787,159 / 96.27%
Unmapped reads	727,099 / 3.73%
Mapped paired reads	18,787,159 / 96.27%
Mapped reads, first in pair	9,397,164 / 48.16%
Mapped reads, second in pair	9,389,995 / 48.12%
Mapped reads, both in pair	18,507,530 / 94.84%
Mapped reads, singletons	279,629 / 1.43%
Secondary alignments	877,136
Read min/max/mean length	20 / 125 / 127.08
Duplicated reads (estimated)	15,724,365 / 80.58%
Duplication rate	69.03%
Clipped reads	937,212 / 4.8%

2.2. ACGT Content

Number/percentage of A's	844,922,071 / 36.93%
Number/percentage of C's	301,931,290 / 13.2%
Number/percentage of T's	827,124,859 / 36.15%
Number/percentage of G's	314,136,015 / 13.73%
Number/percentage of N's	1,711,427,141 / 74.8%
GC Percentage	26.92%

2.3. Coverage

Mean	205.7559
Standard Deviation	979.2465

2.4. Mapping Quality

Mean Mapping Quality	34.69
----------------------	-------

2.5. Insert size

Mean	372.2
Standard Deviation	9,327.82
P25/Median/P75	189 / 219 / 255

2.6. Mismatches and indels

General error rate	0.08%
Mismatches	3,307,987
Insertions	257,360
Mapped reads with at least one insertion	1.18%
Deletions	247,534
Mapped reads with at least one deletion	1.14%
Homopolymer indels	79.6%

SRR15276767

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	22,240,266
Mapped reads	21,622,525 / 97.22%
Unmapped reads	617,741 / 2.78%
Mapped paired reads	21,622,525 / 97.22%
Mapped reads, first in pair	10,816,850 / 48.64%
Mapped reads, second in pair	10,805,675 / 48.59%
Mapped reads, both in pair	21,300,144 / 95.77%
Mapped reads, singletons	322,381 / 1.45%
Secondary alignments	820,356
Read min/max/mean length	20 / 125 / 126.08
Duplicated reads (estimated)	17,518,091 / 78.77%
Duplication rate	59.1%
Clipped reads	1,034,750 / 4.65%

2.2. ACGT Content

Number/percentage of A's	974,413,426 / 37%
Number/percentage of C's	346,262,600 / 13.15%
Number/percentage of T's	951,446,522 / 36.13%
Number/percentage of G's	361,388,283 / 13.72%
Number/percentage of N's	2,166,983,263 / 82.28%
GC Percentage	26.87%

2.3. Coverage

Mean	205.7559
Standard Deviation	979.2465

2.4. Mapping Quality

Mean Mapping Quality	34.69
----------------------	-------

2.5. Insert size

Mean	372.2
Standard Deviation	9,327.82
P25/Median/P75	189 / 219 / 255

2.6. Mismatches and indels

General error rate	0.08%
Mismatches	3,307,987
Insertions	257,360
Mapped reads with at least one insertion	1.18%
Deletions	247,534
Mapped reads with at least one deletion	1.14%
Homopolymer indels	79.6%

SRR15276773

2. Summary

2.1. Globals

Reference size	23,332,839
Number of reads	21,931,766
Mapped reads	21,447,346 / 97.79%
Unmapped reads	484,420 / 2.21%
Mapped paired reads	21,447,346 / 97.79%
Mapped reads, first in pair	10,724,262 / 48.9%
Mapped reads, second in pair	10,723,084 / 48.89%
Mapped reads, both in pair	21,133,636 / 96.36%
Mapped reads, singletons	313,710 / 1.43%
Secondary alignments	750,123
Read min/max/mean length	20 / 125 / 126.19
Duplicated reads (estimated)	17,394,401 / 79.31%
Duplication rate	55.99%
Clipped reads	934,316 / 4.26%

2.2. ACGT Content

Number/percentage of A's	974,445,220 / 37.19%
Number/percentage of C's	341,013,994 / 13.02%
Number/percentage of T's	953,247,286 / 36.38%
Number/percentage of G's	351,259,150 / 13.41%
Number/percentage of N's	2,030,155,628 / 77.49%
GC Percentage	26.42%

2.3. Coverage

Mean	199.3105
Standard Deviation	930.2855

2.4. Mapping Quality

Mean Mapping Quality	34.8
----------------------	------

2.5. Insert size

Mean	363.24
Standard Deviation	9,032.34
P25/Median/P75	195 / 226 / 263

2.6. Mismatches and indels

General error rate	0.07%
Mismatches	3,143,395
Insertions	244,516
Mapped reads with at least one insertion	1.13%
Deletions	238,714
Mapped reads with at least one deletion	1.11%
Homopolymer indels	79.73%

9. Quality check of multiple HiSAT alignments using MultiQC

multiqc .

*here . means the folder containing qualimap results.

MultiQC is a tool that aggregates results from multiple quality control analyses, including BAM file QC reports, into a single interactive report. It helps compare and summarize alignment statistics across different samples in one place. To run multiQC first we have to run qualimap for all the samples.

```
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ multiqc .
/// MultiQC v1.38
version_check | MultiQC Version v1.31 now available!
file_search   | Search path: /home/ibab/NGS/23aug/5_HiSAT2/2_Alignment
searching     | 100% 24/24
qualimap      | Found 4 BamQC reports
write_results | Data      : multiqc_data
write_results | Report   : multiqc_report.html
multiqc       | MultiQC complete
```

```
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment$ cd multiqc_data/
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment/multiqc_data$ ls
BETA-multiqc.parquet  multiqc.log          multiqc_data.json    multiqc_qualimap_bamqc_genome_results.txt
llms-full.txt         multiqc_citations.txt multiqc_general_stats.txt multiqc_sources.txt
```

```
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment/multiqc_data$ cat multiqc_general_stats.txt
Sample qualimap_bamqc-mean_coverage qualimap_bamqc-general_error_rate qualimap_bamqc-percentage_aligned qualimap_bamqc-mapped_reads qual
imap_bamqc-mapped_paired_reads qualimap_bamqc-total_reads qualimap_bamqc-ns qualimap_bamqc-duplication_rate
SRR15276765_sorted 28.9485 0.11 88.63761015330223 4.2891189999999995 4.261018 4.838938 152659450 54.12
SRR15276766_sorted 171.4268 0.08 96.27401154581435 18.787159 18.50753 19.514257999999998 1711427141 69.03
SRR15276767_sorted 205.7559 0.08 97.22242081097411 21.622525 21.300144 22.240266 2166983263 59.1
SRR15276773_sorted 199.3105 0.06999999999999999 97.79124034060914 21.447346 21.133636 21.931766 2030155628 55.9
9
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/5_HiSAT2/2_Alignment/multiqc_data$ cat multiqc_qualimap_bamqc_genome_results.txt
Sample total_reads mapped_reads mapped_paired_reads mapped_bases sequenced_bases duplication_rate mean_insert_size median_inser
t_size mean_mapping_quality ns gc_percentage general_error_rate homopolymer_indels mean_coverage percentage_aligned
SRR15276765_sorted 4838938 4289119 4261018 675450120 522714424 54.12 361.5202 194.0 44.5897 152659450 28.67 0.11 84.7
1 28.9485 88.63761015330223
SRR15276766_sorted 19514258 18787159 18507530 3999872795 2288114235 69.03 332.574 220.0 36.0354 1711427141 26.9
2 0.08 79.79 171.4268 96.27401154581435
SRR15276767_sorted 22240266 21622525 21300144 4800868985 2633510831 59.1 372.1975 219.0 34.685 2166983263 2
6.87 0.08 79.6 205.7559 97.22242081097411
SRR15276773_sorted 21931766 21447346 21133636 4650480742 2619965650 55.99 363.2358 226.0 34.8013 2030155628 2
6.42 0.06999999999999999 79.73 199.3105 97.79124034060914
```



A modular tool to aggregate results from bioinformatics analyses across many samples into a single report.

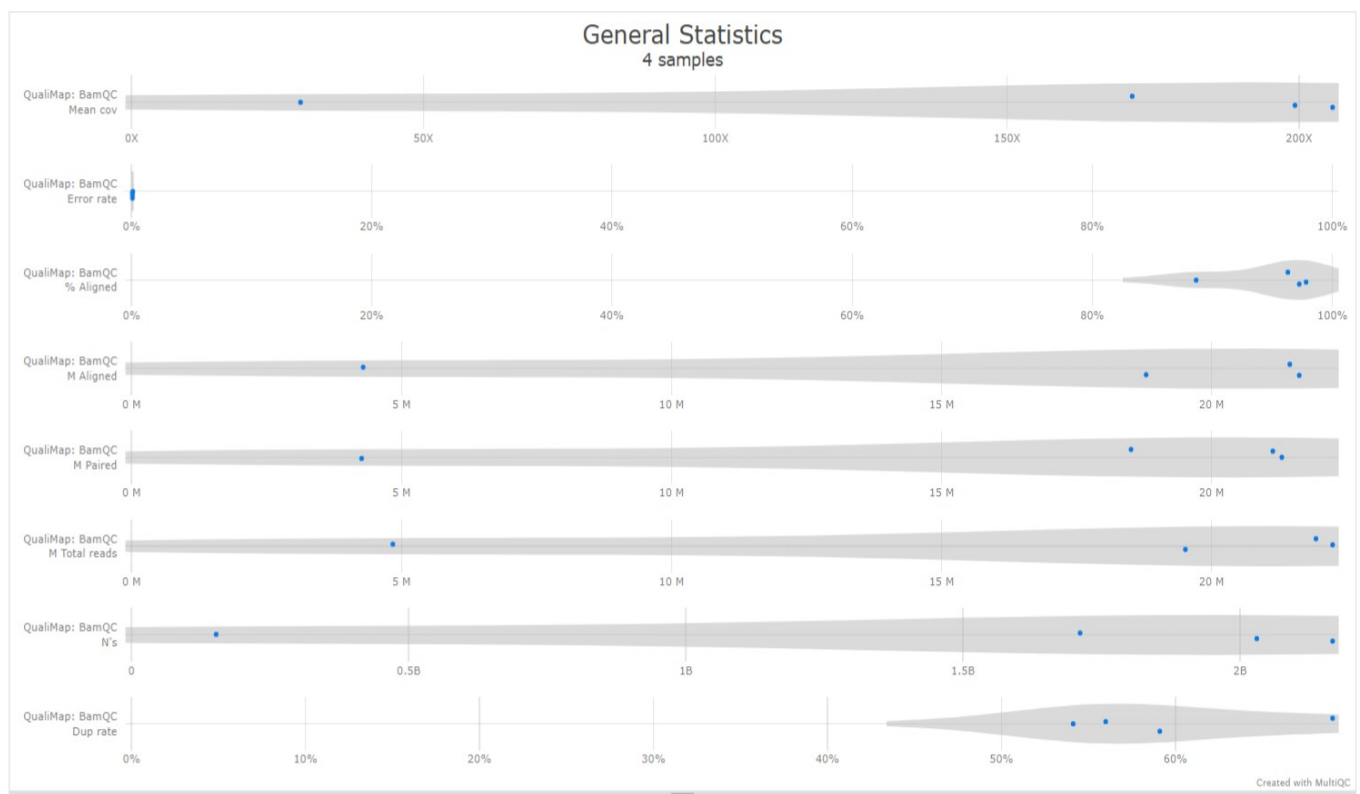
Report generated on 2025-09-18, 06:38 IST based on data in: /home/ibab/NGS/23aug/5_HiSAT2/2_Alignment

Summarize report Copy report prompt

General Statistics

Copied!
Configure columns
Scatter plot
Violin plot
Export as CSV...
Showing 4/4 rows and 8/8 columns.
Copy Prompt
Summarize table

Sample Name	Mean cov	Error rate	% Aligned	M Aligned	M Paired	M Total reads	N's	Dup rate
SRR15276765_sorted	28.9X	0.11 %	88.6 %	4.3M	4.3M	4.8M	152659450	54.1 %
SRR15276766_sorted	171.4X	0.08 %	96.3 %	18.8M	18.5M	19.5M	1711427141	69.0 %
SRR15276767_sorted	205.8X	0.08 %	97.2 %	21.6M	21.3M	22.2M	2166983263	59.1 %
SRR15276773_sorted	199.3X	0.07 %	97.8 %	21.4M	21.1M	21.9M	2030155628	56.0 %



Interpretation: Alignment rates are strong, ranging from 88.6% for SRR15276765 to 97.8% for SRR15276773, while mean coverage is much lower for SRR15276765 (28.95×) compared to the other samples (171–205×), indicating it is under-sequenced. Error rates are low across all samples (0.07–0.11), suggesting accurate alignments. Read counts mirror these trends, with SRR15276765 having fewer mapped reads (~4.29M) than the others (~18–21M). Duplication rates are moderate to high (54–69%), with the highest in SRR15276766, which could slightly reduce effective coverage. The number of ambiguous bases (N's) corresponds with total reads, being lowest in SRR15276765. Overall, the dataset is high quality, but SRR15276765 may require caution due to lower depth, and duplication should be considered in downstream analyses.

Exercise-2: HISAT2 Splice Site and Exon Extraction

A. How can you extract splice site information from a GTF file?

Splice site information can be extracted from a GTF annotation file using the script `hisat2_extract_splice_sites.py`, which comes with the HISAT2 package. This script goes through each transcript defined in the annotation, identifies the positions of exons, and then records the junctions formed between consecutive exons. Each of these junctions represents a donor and acceptor site that RNA splicing uses to remove introns. By running a command such as `hisat2_extract_splice_sites.py Pfalciparum.genome.gtf > splicesites.txt`, we generate a file that lists the coordinates of known splice sites for the reference genome. This file is extremely useful because RNA-seq reads often span across

introns, and without knowledge of where splicing occurs, many reads would be either partially aligned or completely unmapped. By providing HISAT2 with splice site information, the aligner can correctly place reads that cross exon–exon boundaries, improving both accuracy and biological relevance of the alignment.

```
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/exercise_2$ /home/ibab/miniconda3/bin/hisat2_extract_splice_sites.py /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf > splicesites.txt
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/exercise_2$ ls -lh
total 240K
-rw-r--r-- 1 ibab ibab 240K Sep 18 07:16 splicesites.txt
```

PFC10_API_IRAB	13410	13416	+
Pf3D7_01_v3	34761	35887	+
Pf3D7_01_v3	39922	40153	-
Pf3D7_01_v3	43616	43774	-
Pf3D7_01_v3	50415	50571	+
Pf3D7_01_v3	54787	54937	-
Pf3D7_01_v3	59824	60022	+
Pf3D7_01_v3	63184	63346	-
Pf3D7_01_v3	66820	66917	-
Pf3D7_01_v3	70163	70348	-
Pf3D7_01_v3	71691	71802	+
Pf3D7_01_v3	74630	74727	+
Pf3D7_01_v3	82819	83052	-
Pf3D7_01_v3	88002	88108	-
Pf3D7_01_v3	91203	91351	-
Pf3D7_01_v3	99012	99219	+
Pf3D7_01_v3	104774	104927	+
Pf3D7_01_v3	110822	110961	-
Pf3D7_01_v3	111049	111166	-
Pf3D7_01_v3	111244	111337	-
Pf3D7_01_v3	110330	110353	-

- **Column-1: Reference sequence name (chromosome/contig ID)**
This tells us on which chromosome/contig the splice junction occurs.
- **Column-2: Donor site (5' splice site) coordinate:**
This is the genomic position where the intron starts (right after the end of the upstream exon).
- **Column-3: Acceptor site (3' splice site) coordinate**
This is the genomic position where the intron **ends** (just before the downstream exon starts).
- **Column-4: Strand (+ or -)**
Indicates whether the transcript is on the forward (+) or reverse (-) DNA strand.

```
(base) ibab@LAPTOP-BVSTVK8Q:~/NGS/23aug/exercise_2$ wc -l splicesites.txt
8531 splicesites.txt
```

B. How can you extract exon information from a GTF file?

In a similar way, exon coordinates can also be extracted from a GTF file using the HISAT2 script `hisat2_extract_exons.py`. This script scans through the annotation and writes out the start and end positions of every exon in all annotated transcripts, producing a file such as `exons.txt`. Exon information serves as an important anchor for the aligner, because RNA-seq reads are primarily derived from exonic regions. By knowing the precise genomic intervals that correspond to exons, HISAT2 can more confidently map reads that fall entirely within coding regions or untranslated regions of transcripts.

`hisat2_extract_exons.py Pfalciparum.genome.gtf > Pfalciparum.exons.txt`

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/exercise_2$ hisat2_extract_exons.py /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.gtf > Pfalciparum.exons.txt
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/exercise_2$ ls -lh
total 640K
-rw-rw-r-- 1 ibab ibab 397K Sep 18 09:31 Pfalciparum.exons.txt
-rw-rw-r-- 1 ibab ibab 240K Sep 18 09:29 splicesites.txt
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/exercise_2$ wc -l Pfalciparum.exons.txt
14126 Pfalciparum.exons.txt
```

M76611	2	32	+
M76611	33	70	+
M76611	71	124	+
M76611	125	164	+
M76611	203	225	-
M76611	282	388	-
M76611	389	504	-
M76611	505	592	-
M76611	593	623	-
M76611	624	723	-
M76611	733	1485	-
M76611	1515	1629	+

- **Column 1:** Chromosome/Contig name (e.g., M76611)
- **Column 2:** Exon start coordinate (1-based, inclusive)
- **Column 3:** Exon end coordinate (1-based, inclusive)
- **Column 4:** Strand orientation of the exon (+ for forward strand, - for reverse strand)

C. How can you incorporate both splice site and exon information into genome indexing?

Once splice site and exon files have been generated, they can be integrated directly into the HISAT2 genome index during the indexing step. This is done by running a command such as **hisat2-build --ss splicesites.txt --exon Pfalciparum.exons.txt Pfalciparum.genome.fasta Pfalciparum_index**. Supplying both files ensures that HISAT2 is “annotation-aware,” meaning it recognizes known exon boundaries and splice junctions during alignment. This has several advantages: reads spanning known junctions are mapped more accurately, ambiguous alignments are reduced, and overall mapping rates are improved. At the same time, HISAT2 is still capable of discovering novel splice sites if they exist in the data, so incorporating annotation does not limit discovery. This combination of genome sequence and annotation allows the index to capture both the structure and function of genes, making downstream analyses such as transcript assembly, differential expression, or isoform discovery more reliable and biologically meaningful.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/23aug/exercise_2$ hisat2-build --ss splicesites.txt --exon Pfalciparum.exons.txt /home/ibab/NGS/23aug/2_Reference/Pfalciparum.genome.fasta Pfalciparum_index
Settings:
  Output files: "Pfalciparum_index.*.ht2"
  Line rate: 7 (line is 128 bytes)
```